

Project Overview

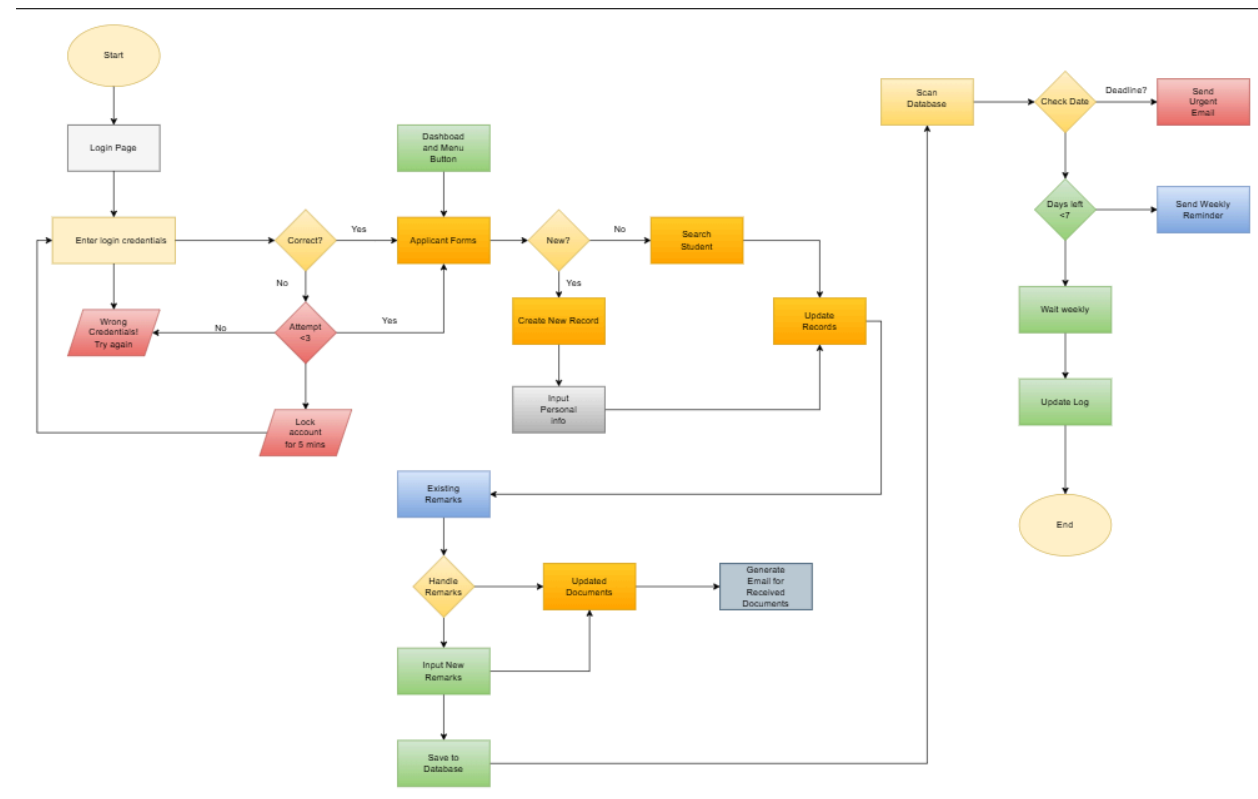
The PUPOUS Document Tracking System (PDTS) is a full-stack, on-premises platform built exclusively for the Polytechnic University of the Philippines Open University System Registrar's Office. Its core mission is to digitize, centralize, and automate the complete lifecycle management of student and graduating documents — replacing fragmented manual tracking with a single, secure, and auditable digital system.

PDTS handles documents across all five curriculum types recognized by PUP: College Undergraduate, Technical-Vocational Education and Training (TVET). The system automates tracking number generation, status-driven workflow (Pending → Received / Rejected / Resubmitted), real-time email notifications to students, and a comprehensive timestamped activity log — all enforced through Role-Based Access Control (RBAC) that limits CRUD operations exclusively to the Registrar and DB Admin.

Core Features at a Glance

Feature	Description
Document Upload & Intake	Registrar uploads scanned documents; system auto-assigns unique tracking numbers (DOC-YYYY-NNNNN) and sets status to Pending.
Status Workflow	Four statuses: Pending → Received or Rejected. Rejected documents enter a Resubmitted → Pending re-review cycle.
Automated Email Notifications	Status changes (Receive, Reject, Resubmit) trigger instant emails to students, including rejection reasons.
Role-Based Access Control	Only Registrar and DB Admin can create, update, or manage data. Student profiles can never be deleted.
Advanced Filter & Search	Dashboard filters by document status, location, enrollment category, and curriculum type.
Immutable Activity Log	Every action is timestamped and stored in an append-only audit log — no update or delete permitted.

Business Process Architecture



Login and Access Control

- Start → Login Page → Enter credentials
- If credentials are correct, the user proceeds to the dashboard.
- If credentials are wrong, the system allows up to 3 attempts before temporarily locking the account for 5 minutes.

Applicant Records

- From the Dashboard, the user can access Applicant Forms.
- If the applicant is new, the system creates a new record and collects personal information.
- If the applicant already exists, the system allows searching for the student and updating records.

Remarks & Documentation

- Existing remarks can be viewed and managed.
- New remarks can be added and saved to the database.
- Updated documents trigger an email notification confirming receipt.

Database Monitoring & Deadlines

- The system periodically scans the database and checks deadlines.
- If a deadline is imminent, it sends an urgent email.
- If there are less than 7 days left, it sends a weekly reminder.
- This reminder process repeats weekly until resolved.

Logging & Completion

- Every action updates the system log.
- The process concludes at the End node.

Programming Language and Technologies

Frontend:	HTML5 · CSS3 · JavaScript
Backend:	Java (Spring Boot) · JavaMail
Database:	MySQL · MySQL Workbench

The PUPOUS Document Tracking System will use HTML, CSS, and JavaScript for the frontend, while Java and Spring Boot will be used for the backend to process records, manage user access, and connect to the database (Balter, 2023).

User Levels

#	USER LEVEL	FEATURES
1	Staff (Limited Access)	- Input applicant personal data - Submit applications into the system - View submissions and records - Upload and update documents - Forward remarks for registrar validation - Trigger notifications (but not approve them) - Limited access to logs (view-only)
2	Registrar (Full Access)	- Manage applicant records (create, update, delete) - Validate and approve documents - Add and handle remarks - Generate and send notifications (urgent emails, weekly reminders) - Monitor deadlines and compliance - Access system logs and audit trail

Interface/Flow

This shows the detailed screen layout of the application as well as the actual flow. Each interface shall have a short discussion below the image. Use the Figure Number of each UI layout.

Figure 1. Login Page

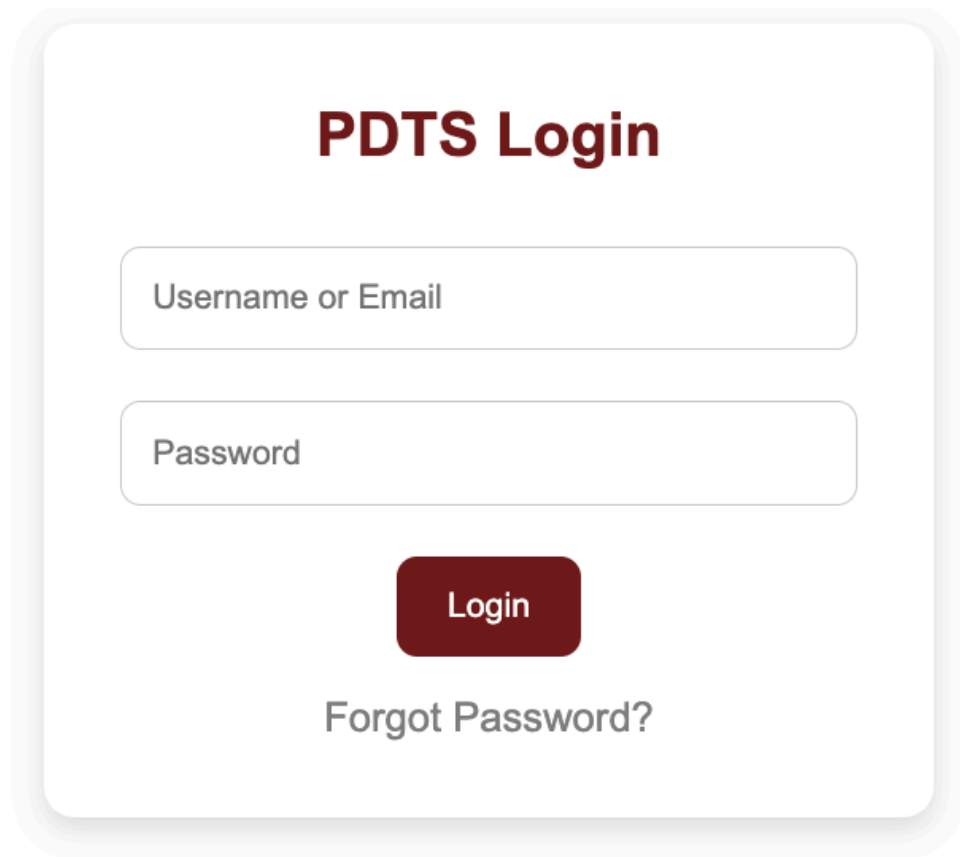
The image shows a login interface for 'PDTS Login'. It features a white rounded rectangle with a subtle drop shadow on a light gray background. Inside the rectangle, the title 'PDTS Login' is centered at the top in a bold, dark red font. Below the title are two input fields: the first is labeled 'Username or Email' and the second is labeled 'Password', both in a light gray font. Below these fields is a dark red rectangular button with the word 'Login' in white. At the bottom of the rectangle, the text 'Forgot Password?' is displayed in a light gray font.

Figure 1

The Login Page of the PUPOUS Document Tracking System. It contains the username and password fields together with the Login button. This interface is used by Staff, Registrar, and DB Admin to access the system securely.

Figure 2. Dashboard

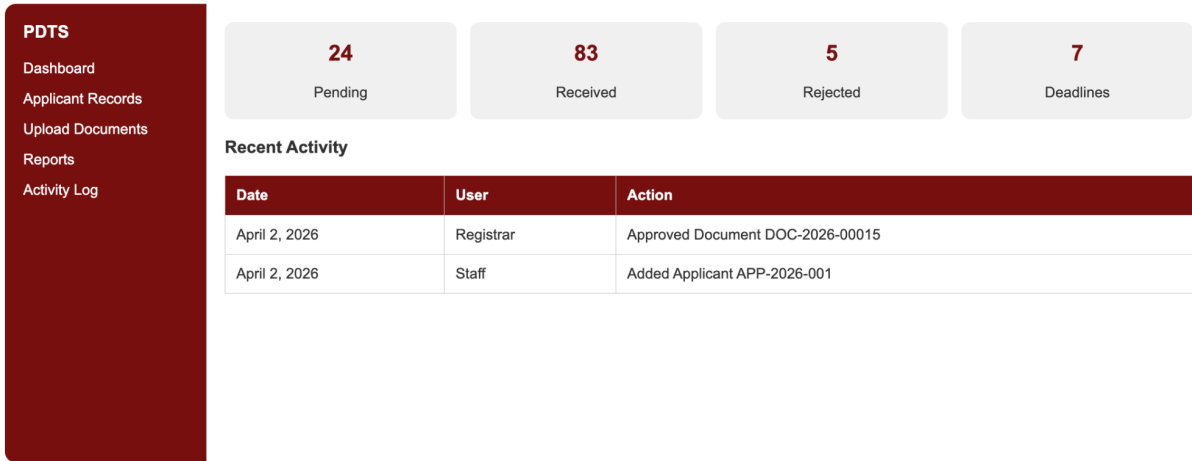


Figure 2

The Dashboard of the system. It displays the number of pending, received, and rejected documents, together with deadlines and recent activities. This interface helps users quickly monitor the status of all submitted documents.

Figure 3. Applicant Records Page

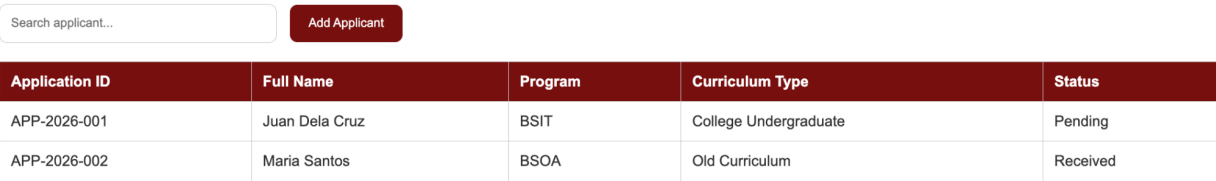


Figure 3

The Applicant Records Page. It contains the list of students together with their application ID, program, curriculum type, and document status. The user may search, add, or update applicant records in this interface.

Figure 4. Add/Edit Applicant Form



The form consists of several input fields stacked vertically. The first field is labeled 'Full Name' and has a red eye icon on the right. The second field is labeled 'Application ID'. The third field is labeled 'Email Address'. The fourth field is labeled 'Program'. The fifth field is a dropdown menu labeled 'College Undergraduate' with a downward arrow. The sixth field is labeled 'Contact Number'. Below the input fields is a red button labeled 'Save Applicant'.

Figure 4

The Add/Edit Applicant Form. This interface allows the user to enter or update the student's personal information, program, and curriculum type before submitting documents.

Figure 5. Document Upload Page

Transcript of Records

Choose File | No file chosen

Remarks

DOC-2026-00015

Submit Document

Figure 5

The Document Upload Page where users may upload student requirements. After uploading the file, the system automatically generates a tracking number and changes the status to Pending.

Figure 6. Document Status Flow

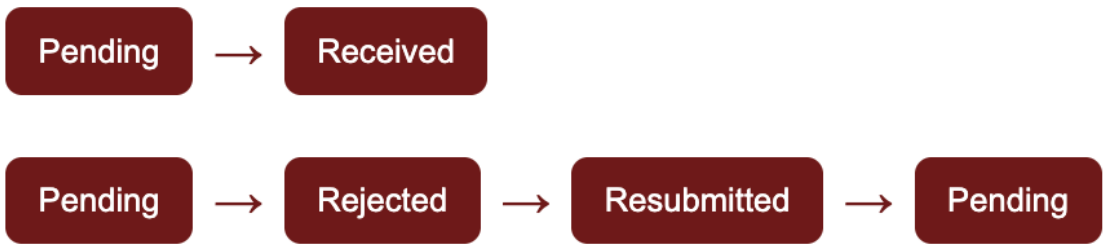


Figure 6

The document status process of the system. Submitted documents may either be marked as Received or Rejected. Rejected documents may be resubmitted and reviewed again.

Figure 7. Email Notification Page

Your submitted Transcript of Records has been marked as Rejected due to incomplete information. Please resubmit within 7 days.

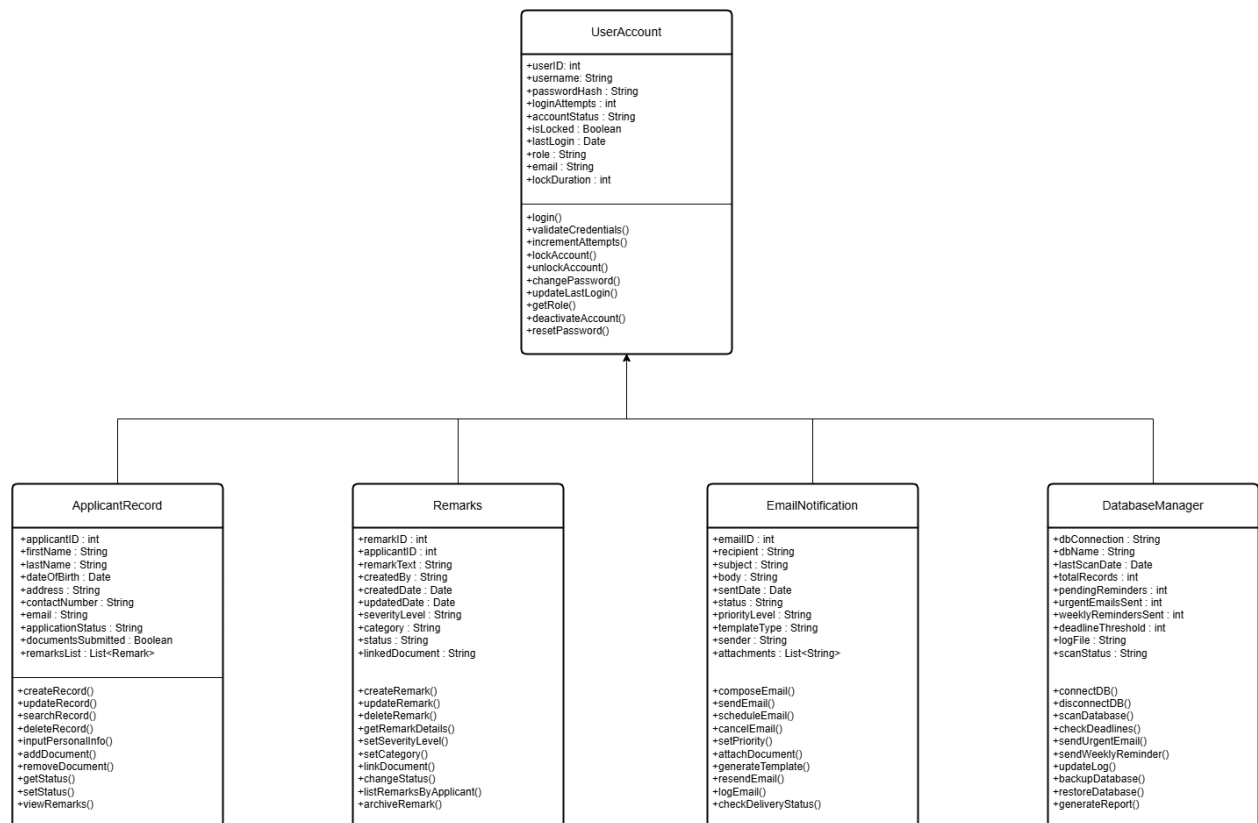
Figure 7
The Email Notification Page. This interface sends automatic messages to students whenever there is an update in the status of their submitted documents.

Figure 8. Activity Log Page

Date and Time	User	Action	Applicant
April 2, 2026 10:15 AM	Registrar	Approved Document	Juan Dela Cruz
April 2, 2026 11:30 AM	Staff	Added Applicant Record	Maria Santos

Figure 8
The Activity Log Page. It records all actions performed in the system, including login, upload, approval, and updates. This helps maintain security and accountability within the system.

Class and Attributes in UML Diagram



UserAccount – handles login, credentials, and account status

ApplicantRecord – manages student data and application status

Remark – tracks comments and feedback on student records

EmailNotification – generates and sends emails based on workflow triggers

DatabaseManager – scans records, checks deadlines, and sends reminders

Reports

This lists all the reports that the project can generate. An example of a report is a Sales Report – this shows daily, monthly, yearly, or from a particular date, the sales of a particular product. At least 5 reports

#	REPORT	DESCRIPTION
1	Document Submission Report	Shows all submitted documents, including application ID, student name, document type, date submitted, and status. The report may be generated daily, weekly, monthly, or from a selected date range.
2	Pending Documents Report	Shows all documents that are still pending review. The report may be filtered by date, academic program, curriculum type, or student category.
3	Rejected and Resubmitted Report	Shows all rejected documents together with the reason for rejection and whether the student has already resubmitted the requirement. This may be generated weekly, monthly, or by selected date range.
4	Deadline Monitoring Report	Shows students with upcoming or overdue submission deadlines. The report may be viewed for the next 7 days, current month, or overdue list.
5	Student Document Status Report	Shows the complete document history of a specific student, including all uploaded requirements and their status changes from submission to approval.
6	Activity Log Report	Shows all system activities such as login, upload, approval, rejection, and update of records. The report may be generated daily, weekly, or monthly.

Report 1. Document Submission Report

Application ID	Student Name	Program	Document Type	Date Submitted	Status
APP-2026-001	Juan Dela Cruz	BSIT	Transcript of Records	April 1, 2026	Pending
APP-2026-002	Maria Santos	BSOA	Birth Certificate	April 2, 2026	Received
APP-2026-003	Carlo Reyes	BSOD	Diploma	April 2, 2026	Rejected

Project Staff

This discusses the assignment of the team members. Please ensure that all team members have duties and responsibilities, and this will be a basis for rating objectively.

#	NAME	ROLE	RESPONSIBILITIES
	San Pedro, Jonathan		

Project Schedule

This shall discuss the schedule of the development of the project.

#	TASK	START	END	DURATION	MEMBER
1	Project overview	March 6, 2026	March 10, 2026	5	Balakwit

2	Business Process Architecture	March 11, 2026	March 13, 2026	3	Cioco
3	Programming Language and Technologies	March 14, 2026	March 16, 2026	3	Gravoso
4	User Levels	March 17, 2026	March 18, 2026	2	Balakwit
5	Interface Flow	March 19, 2026	March 21, 2026	3	Pamintuan
6	Class and Attributes in UML Diagram	March 22, 2026	March 24, 2026	3	San Pedro
7	Reports	March 25, 2026	March 26, 2026	2	Pamintuan
8	Project Budget	March 27, 2026	March 28, 2026	2	San Pedro
9	Risk	March 29, 2026	April 6 30, 2026	9	Cioco
10	References	April 8, 2026	April 10, 2026	3	Gravoso

Project Budget

This discusses the cost of the project.

#	ITEM	DESCRIPTION	COST
	Software Development	Coding of basic document tracker (CRUD functions, login,	5,000

	Database Setup	Free/open-source DB (MySQL/PostgreSQL) setup and schema design	0
	UI/UX Design	Simple interface using free CSS frameworks (Bootstrap, Tailwind)	2,000
	Testing & QA	Manual testing and bug fixing	1,000
	Documentation	Project report, user manual, and presentation materials	1,500
	Hosting	Student-friendly cloud hosting (Heroku free tier / local server)	0
	Miscellaneous	Printing, materials, and contingency	
		TOTAL:	PHP 10,500 – 12,000

Risks

This lists all the risks of problems that may be encountered and how to solve them.

#	RISK	MITIGATION
1	Loss of internet connection or server failure may prevent users from accessing the system.	Maintain a local backup of the database and allow the system to save data automatically once the connection is restored.
2	Incorrect or incomplete student information may be entered into the system.	Add validation rules, required fields, and a confirmation message before saving the record.

3	Unauthorized users may gain access to confidential student records.	Use secure login credentials, role-based access control, and encrypted passwords.
4	Duplicate applicant records or duplicate document uploads may occur.	Automatically generate unique application IDs and tracking numbers, and check existing records before saving.
5	Email notifications may fail to send because of network issues or wrong email addresses.	Add an email retry function and require users to verify the student's email address before sending notifications.
6	Uploaded files may be lost or corrupted during submission.	Store files in a secure backup folder and allow users to re-upload documents if an error occurs.
7	The system may become slow when many records are stored.	Optimize the database, archive old records, and use search and filter functions to improve performance.
8	Users may accidentally delete or modify important student records.	Limit delete and edit permissions only to the Registrar or DB Admin and keep an activity log for all actions.
9	Students may miss submission deadlines because they are not informed.	Send automatic reminder emails and display approaching deadlines on the dashboard.
10	Power interruption may interrupt ongoing work and cause unsaved data loss.	Enable automatic saving and use an uninterruptible power supply (UPS) for the server or computer.

References

This lists all the references in the 7th edition APA.

Prepared by:

Signature of the Team Leader	
Name of the Team Leader	